

1 The density matrix, entanglement entropy and their connection to singular value decomposition (SVD)

1.1 Main goal

Prove that for λ_i the eigenvalues of the reduced density matrix and s_i the Schmidt values,

$$S_{\text{Neum}} \equiv \text{Tr}(\rho_A \log \rho_A) = \sum_i \lambda_i \log \lambda_i = \sum_i s_i^2 \log s_i^2. \quad (1)$$

1.2 The density operator

Let $\{p_i, |\psi_i\rangle\}$ be an ensemble of pure states, describing a system that is in one of the states $|\psi_i\rangle$ with probability p_i , i.e. the system is not fully known. Define the density operator as

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \quad p_i \in \mathbb{R} \ \& \ \sum_i p_i = 1, \quad (2)$$

which is a positive semi-definite operator of trace one (see for example Nielsen & Chuang).

1.3 Observables on a composite system

Take a system A with some observable M , and a system B , giving the composite system AB (see Box 2.6). Assume we can measure M on system A ; then a corresponding measurement on AB is $\tilde{M} = M \otimes \mathbb{1}_B$.

Proof. Take AB to be prepared in $|m\rangle|\psi\rangle$ with $M|m\rangle = m|m\rangle$. Applying the measurement to $|m\rangle|\psi\rangle$ should yield m with probability one, i.e. the corresponding projector for $\tilde{M}$ is $P_m \otimes \mathbb{1}_B$. By the spectral theorem, $\tilde{M} = \sum_m m P_m \otimes \mathbb{1}_B = M \otimes \mathbb{1}_B$ for a general state of AB . $\square$

1.4 The reduced density matrix as a partial trace

Measurement averages must be the same whether computed via the full ρ^{AB} or via some ρ^A associated to system A , i.e.

$$\text{tr}(M\rho^A) \stackrel{!}{=} \text{tr}(\tilde{M}\rho^{AB}) = \text{tr}((M \otimes \mathbb{1}_B)\rho^{AB}). \quad (3)$$

Take $\rho^A = \text{tr}_B(\rho^{AB})$ where

$$\text{tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) \equiv |a_1\rangle\langle a_2| \text{tr}(|b_1\rangle\langle b_2|) \quad \& \quad \text{tr}_B(\cdot) \text{ linear.} \quad (4)$$

Write ρ^{AB} by using the eigenbasis of M for system A with $M|i\rangle = m_i|i\rangle$ and $\langle i|j\rangle = \delta_{ij}$:

$$\rho^{AB} = \sum_\alpha p^\alpha |\psi\rangle\langle\psi|^\alpha = \sum_\alpha p^\alpha \sum_{ijkl} c_{ij}^\alpha (c_{k\ell}^\alpha)^* |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A, \quad (5)$$

$$\rho^A = \sum_\alpha \sum_{ijkl} p^\alpha c_{ij}^\alpha (c_{k\ell}^\alpha)^* |i\rangle\langle k|_A \underbrace{\text{tr}(|j\rangle\langle\ell|_B)}_{=\delta_{j\ell}} = \sum_{\alpha, ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* |i\rangle\langle k|_A. \quad (6)$$

Therefore

$$\text{tr}(M\rho^A) = \sum_{\alpha,ijk} p^\alpha c_{ij}^\alpha (c_{kj}^\alpha)^* \underbrace{\text{tr}(m_i |i\rangle\langle k|_A)}_{=m_i\delta_{ik}} = \sum_{\alpha,ij} p^\alpha m_i c_{ij}^\alpha (c_{ij}^\alpha)^*, \quad (7)$$

and similarly $\text{tr}((M \otimes \mathbb{1}_B)\rho^{AB}) = \sum_{\alpha,ij} p^\alpha m_i c_{ij}^\alpha (c_{ij}^\alpha)^*$, which proves the claim. One can also easily show that this is the only map that conserves consistency (see Box 2.6).

1.5 Reduced density matrix of a pure state

Now examine the reduced density matrix for a pure state,

$$\rho_{AB} = |\psi\rangle\langle\psi| = \sum_{ijk\ell} c_{ij}(c_{k\ell})^* |i\rangle_A |j\rangle_B \langle\ell|_B \langle k|_A, \quad (8)$$

$$\rho_A = \text{tr}_B(\rho_{AB}) = \sum_{ijk} c_{ij}(c_{kj})^* |i\rangle\langle k|_A, \quad (9)$$

such that the matrix representation is

$$\boxed{\rho_A = \psi\psi^\dagger}, \quad (10)$$

using that the c_{ij} parametrize the matrix $\psi \in \mathbb{C}^{d_A \times d_B}$, i.e. $|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B$.

1.6 Connection to the SVD

Now use the SVD on ψ :

$$\psi = USV^\dagger \quad \text{with } U \in \mathbb{C}^{d_A \times d_A}, V \in \mathbb{C}^{d_B \times d_B} \text{ unitary.} \quad (11)$$

Then, using $V^\dagger V = \mathbb{1}$,

$$\rho_A = USV^\dagger V S U^\dagger = US^2 U^\dagger, \quad S^2 = \text{diag}(s_1^2, s_2^2, \dots) \in \mathbb{R}_{\geq 0}^{d_A \times d_A}. \quad (12)$$

Furthermore, since ρ_A is hermitian we can diagonalize it with $\rho_A = O D O^\dagger = U S^2 U^\dagger$ and $D = \text{diag}(\lambda_1, \dots)$, so that

$$\boxed{\lambda_i = s_i^2}, \quad (13)$$

i.e. the eigenvalues are given by the (squared) Schmidt values.

1.7 Functions of the density matrix via the spectral theorem

Since ρ_A is hermitian, we can use the spectral theorem to write it as $\rho_A = \sum_i \lambda_i P_i$ where λ_i are its eigenvalues and the eigenvectors $|u_i\rangle$ give $P_i = |u_i\rangle\langle u_i|$, since

$$\rho_A = U D U^\dagger = (|u_1\rangle, \dots, |u_n\rangle) \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \begin{pmatrix} \langle u_1| \\ \vdots \\ \langle u_n| \end{pmatrix}, \quad \langle u_i | u_j \rangle = \delta_{ij}. \quad (14)$$

Thus a function $f(\cdot)$ applied to ρ_A gives

$$\begin{aligned} f(\rho_A) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \rho_A^n \stackrel{(*)}{=} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \sum_i \lambda_i^n |u_i\rangle\langle u_i| \\ &= \sum_i \left(\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} \lambda_i^n \right) |u_i\rangle\langle u_i| = \sum_i f(\lambda_i) |u_i\rangle\langle u_i| = U \text{diag}(f(\lambda_i)) U^\dagger, \end{aligned} \quad (15)$$

where $(*)$ uses $\rho_A^n = \sum_i \lambda_i^n |u_i\rangle\langle u_i|$ (e.g. $\rho_A \rho_A = \sum_i \lambda_i^2 |u_i\rangle\langle u_i|$).

1.8 The Von Neumann entanglement entropy

Therefore, for the Von Neumann entanglement entropy,

$$\begin{aligned} S &= \text{Tr}(\rho_A \log \rho_A) = \text{Tr}(\rho_A U \log(\text{diag}(\lambda_i)) U^\dagger) \\ &= \text{Tr}(U \text{diag}(\lambda_i) \underbrace{U^\dagger U}_{=1} \log(\text{diag}(\lambda_i)) U^\dagger) \\ &= \sum_i \lambda_i \log \lambda_i = \boxed{\sum_i s_i^2 \log s_i^2}. \end{aligned} \quad (16)$$

S is zero for two unentangled systems and non-zero for entangled systems.

1.9 Example

In our case, for example, $L = 2$, $N_f = 1$, $N_{\text{ph}} = 1$:

$$\{|01\rangle, |10\rangle\} \text{ \& \; } \{|0\rangle, |1\rangle\} \text{ as bases of } \mathcal{H}_f \text{ \& \; } \mathcal{H}_{\text{ph}}, \quad (17)$$

$$\Rightarrow \{|01, 0\rangle; |01, 1\rangle; |10, 0\rangle; |10, 1\rangle\} \text{ as basis of } \mathcal{H}_f \otimes \mathcal{H}_{\text{ph}}, \quad (18)$$

$$\Rightarrow |\psi\rangle = \sum_{i=1}^2 \sum_{j=1}^2 \psi_{ij} |i\rangle_f |j\rangle \text{ with } (\psi)_{ij} = \psi_{ij} \text{ \& \; } \psi \in \mathbb{C}^{2 \times 2}, \quad (19)$$

$$\Rightarrow S(\rho_A) = \sum_{\ell} s_{\ell}^2 \log s_{\ell}^2. \quad (20)$$

Note that since $\text{tr} \rho^{AB} = \text{tr} \rho^A = \text{tr} \rho^B = 1$ we must have $\sum_{\ell} \lambda_{\ell} = \sum_{\ell} s_{\ell}^2 = 1$.

1.10 References

- M. Nielsen & I. Chuang, *Quantum Computation and Quantum Information* (density operator, partial trace; Box 2.6).
- U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, beginning of Chapter 4 (MPS).